

# Quintic Modification to Lifshitz Quasi-topological Black Holes

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## Abstract

We extend the analysis of Lifshitz black holes to quintic order in five-dimensional quasi-topological gravity coupled to a massive Abelian vector field. Starting from a static ansatz with a constant-curvature horizon, we derive the reduced field equations and identify the radially conserved quantity of the one-dimensional effective system. We then analyze the algebraic conditions that permit Lifshitz backgrounds, both in the absence and in the presence of the massive vector field. Since closed-form black-hole solutions are not available for the generic quintic theory, we construct numerical solutions using near-horizon expansions and a shooting method. We present solutions for the relativistic branch  $z = 1$  and the Lifshitz branch  $z = 2$ , covering the three horizon topologies  $k = -1, 0, +1$ . The numerical profiles of the metric functions and the gauge-field function show behavior that is qualitatively consistent with earlier studies of cubic and quartic quasi-topological Lifshitz black holes. We also compute the Wald entropy and Hawking temperature, and examine the local thermal behavior through logarithmic entropy – temperature plots. For the representative parameter choices considered here, the numerical branches shown possess positive heat capacity.

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# 1 Introduction

Gauge/gravity duality offers a powerful framework in which strongly coupled quantum systems can be studied via classical gravitational dynamics in one higher dimension [1–3]. Its most thoroughly understood realization is the AdS/CFT correspondence, but holographic techniques have also been generalized to systems whose scaling behavior departs from relativistic invariance [4, 5]. A particularly prominent class of such nonrelativistic systems is characterized by Lifshitz scaling,

$$t \rightarrow \lambda^z t, \quad x_i \rightarrow \lambda x_i, \quad (1)$$

where  $z$  denotes the dynamical critical exponent. For  $z = 1$  the scaling reduces to the relativistic symmetry familiar from conformal field theories, whereas  $z \neq 1$  encodes an anisotropic scaling between time and space. Gravitational geometries with Lifshitz asymptotics therefore provide a natural setting for holographic models of non-relativistic critical phenomena and related condensed-matter systems [6–10].

From a gravitational standpoint, it is equally natural to incorporate higher-curvature corrections to the Einstein-Hilbert action. These corrections emerge in effective descriptions of gravity and extend the space of gravitational couplings accessible within holographic models [11, 12]. This is significant because pure Einstein gravity contains only a restricted set of parameters and therefore cannot capture the full range of possible dual field theories, especially when quantities such as central charges, transport coefficients, and thermodynamic response functions are permitted to vary independently [13, 14]. Lovelock gravity is particularly well suited to this context because its higher-curvature densities yield field equations that contain at most second derivatives of the metric [15]. In a fixed low dimension, however, the higher-order Lovelock terms either vanish identically or become topological, which limits their usefulness for five-dimensional holographic applications.

Quasi-topological gravity provides a means to preserve some of the appealing features of Lovelock theory while still yielding nontrivial higher-curvature dynamics in five dimensions [17, 18]. Quasi-topological interactions are specifically designed so that the field equations simplify considerably on highly symmetric backgrounds—including static black-hole ansätze—despite the fact that the theory is not second order for a completely generic metric. This property makes quasi-topological gravity a particularly useful laboratory for investigating how additional curvature couplings modify black-hole solutions and their holographic interpretation [13, 14].

Asymptotically Lifshitz black holes in quasi-topological gravity have previously been investigated at cubic and quartic order [16, 19]. The cubic theory showed that quasi-topological interactions can support Lifshitz black holes in five dimensions, and that the resulting solutions can be constructed numerically by combining near-horizon expansions with shooting methods. The quartic extension showed that this structure continues to hold when an additional higher-curvature coupling is introduced; the quartic term alters the black-hole profiles and thermodynamic behavior but preserves the overall pattern seen in the lower-order solutions. Quintic quasi-topological gravity, a higher-order extension of this framework, has been constructed more recently. Black-hole solutions in the quintic theory have begun to be investigated in a variety of other settings, including static higher-dimensional solutions, thermodynamic analyses, boundary terms, conserved quantities, and charged black-hole geometries [20–23]. These advances provide strong motivation to study Lifshitz black holes within the quintic theory.

In this paper, we push this framework to quintic order—the highest nontrivial curvature order that can be consistently constructed in five dimensions via the quasi-topological recipe. This extension is not merely a formal exercise; it introduces an additional coupling that significantly enriches the gravitational dynamics while maintaining the simplifying property of the reduced field equations on static, constant-curvature backgrounds. We derive the radially conserved quantity of the one-dimensional effective system, which generalizes the lower-order expressions and now involves all couplings up to quintic order in a nontrivial way.

We analyze Lifshitz backgrounds first without and then with the massive vector field. In the absence of matter, the requirement of an exact Lifshitz background imposes algebraic constraints among the cosmological constant and the quasi-topological couplings. In the matter-supported case, the charge and mass of the vector field are determined by the dynamical exponent and the curvature couplings. Because closed-form black-hole solutions are not generally available for  $z \neq 1$ , we construct the solutions numerically. The near-horizon expansion provides the initial data, and a shooting method is employed to enforce the asymptotically Lifshitz boundary conditions at large radius.

We perform the numerical analysis for  $z = 1$  and  $z = 2$ , covering the three horizon topologies  $k = -1, 0, +1$ . We examine the behavior of the metric function  $f(r)$  for both values of  $z$ , compare  $f(r)$  and  $g(r)$  for the  $z = 2$  case, and study the full set of functions  $f(r)$ ,  $g(r)$ , and  $h(r)$  for asymptotically Lifshitz black holes with  $z = 2$ . We also compute the thermodynamic quantities and employ logarithmic plots of the entropy and temperature to probe the local thermodynamic behavior. The qualitative features of the quintic solutions are consistent with the pattern seen in the cubic and quartic quasi-topological Lifshitz black holes: the quintic interaction expands the coupling space and alters the numerical profiles, while the overall structure of the solutions stays close to that of the lower-order theories.

The remainder of the paper is structured as follows. In Section 2 we present the quintic quasi-topological action, the Lifshitz ansatz, and the reduced field equations. Section 3 is devoted to the derivation of the conserved quantity associated with radial evolution. In Section 4 we discuss Lifshitz solutions both with and without matter. Section 5 builds the near-horizon and asymptotic expansions used in the numerical analysis, and Section 6 presents the numerical black-hole solutions for  $z = 1$  and  $z = 2$ . In Section 7 we compute the entropy and temperature and discuss the local thermodynamic behavior of the solutions. Finally, Section 8 gives our concluding remarks.

## 2 Quintic quasi-topological gravity and Lifshitz ansatz

### 2.1 Brief review of quasitopological gravity

We study quasi-topological gravity including curvature terms up to fifth order, coupled to a massive Abelian vector field. The action is given by

$$I = \int d^{n+1}x \sqrt{-g} \left[ -2\Lambda + \mathcal{L}_1 + \mu_2 \mathcal{L}_2 + \mu_3 \mathcal{L}_3 + \mu_4 \mathcal{L}_4 + \mu_5 \mathcal{L}_5 - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2} m^2 A_\mu A^\mu \right], \quad (2)$$

where  $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ . We omit the overall gravitational factor from the action and work in units with  $16\pi G_{n+1} = 1$ , except where explicitly noted otherwise. The first two gravitational densities read

$$\mathcal{L}_1 = R, \quad \mathcal{L}_2 = R_{abcd} R^{abcd} - 4R_{ab} R^{ab} + R^2, \quad (3)$$

and  $\mathcal{L}_3$ ,  $\mathcal{L}_4$ ,  $\mathcal{L}_5$  are the cubic, quartic, and quintic quasi-topological densities, respectively. The cubic density is given by

$$\begin{aligned} \mathcal{L}_3 = & R_a^c R_b^d R_c^e R_d^f R_e^a R_f^b + \frac{1}{(2n-1)(n-3)} \left[ \frac{3(3n-5)}{8} R_{abcd} R^{abcd} R \right. \\ & - 3(n-1) R_{abcd} R^{abc} R^{de} + 3(n+1) R_{abcd} R^{ac} R^{bd} + 6(n-1) R_a^b R_b^c R_c^a \\ & \left. - \frac{3(3n-1)}{2} R_a^b R_b^a R + \frac{3(n+1)}{8} R^3 \right]. \end{aligned} \quad (4)$$

The explicit forms of the quartic and quintic densities are rather lengthy; here it suffices to note that they can be expressed as linear combinations of independent curvature invariants,

$$\mathcal{L}_4 = \sum_{i=1}^{14} b_i \mathcal{R}_i^{(4)}, \quad \mathcal{L}_5 = \sum_{i=1}^{24} c_i \mathcal{R}_i^{(5)}, \quad (5)$$

with the coefficients  $b_i$  and  $c_i$  chosen so that the field equations simplify for the static constant-curvature ansatz we consider below. For the quartic terms we adopt the coefficients of quartic quasi-topological gravity; the quintic density is taken from the standard representative construction of quintic quasi-topological gravity [14, 20]. At quintic order, the requirement that the field equations simplify leaves one free overall choice in the coefficients. Here we adopt the standard representative quintic quasi-topological density given in Ref. [20]; the basis of curvature invariants and our conventions are collected in Appendix A.

The dimensionless couplings  $\hat{\mu}_i$  appearing below are the normalized combinations of the bare couplings  $\mu_i$  that emerge upon substituting the Lifshitz ansatz into the action. The normalization is fixed by requiring that the reduced action become the polynomial in  $\Psi$  shown in Eq. (10); throughout the analytic formulas we keep the hats to distinguish these combinations from the bare coefficients in Eq. (2).

## 2.2 Lifshitz ansatz and reduced action

We adopt the static Lifshitz ansatz

$$ds^2 = -\frac{r^{2z}}{L^{2z}} f(r) dt^2 + \frac{L^2 dr^2}{r^2 g(r)} + r^2 d\Omega_{k,n-1}^2, \quad (6)$$

in which  $z$  is the Lifshitz dynamical exponent. The metric  $d\Omega_{k,n-1}^2$  denotes an  $(n-1)$ -dimensional space of constant curvature  $(n-1)(n-2)k$ , where  $k = -1, 0, +1$  correspond, respectively, to hyperbolic, planar, and spherical horizon topologies. For  $k \neq 0$ , the constant-curvature metric can be written explicitly as

$$d\Omega_{k,n-1}^2 = d\theta_1^2 + k^{-1} \sin^2(\sqrt{k} \theta_1) \left[ d\theta_2^2 + \sum_{i=3}^{n-1} \left( \prod_{j=2}^{i-1} \sin^2 \theta_j \right) d\theta_i^2 \right], \quad (7)$$

and the planar case corresponds to the appropriate  $k \rightarrow 0$  limit. The boundary conditions for an asymptotically Lifshitz spacetime are

$$\lim_{r \rightarrow \infty} f(r) = 1, \quad \lim_{r \rightarrow \infty} g(r) = 1. \quad (8)$$

We take the massive vector field to be compatible with the symmetries of the metric. Accordingly, we take the vector field to be of the form

$$A = A_t(r) dt, \quad A_t(r) = q \frac{r^z}{L^z} h(r), \quad (9)$$

where  $q$  is a constant and, for the matter-supported Lifshitz branch,  $h(r) \rightarrow 1$  at infinity.

In what follows we work exclusively in a five-dimensional bulk, setting  $n = 4$ , which is the natural dimension for quasi-topological constructions whose dual boundary theory is four-dimensional. After inserting the ansätze (6) and (9) into the action and dropping the constant transverse volume factor, the reduced action can be written in the compact form

$$I_{\text{red}} \propto \int dr r^{z-1} \sqrt{\frac{f}{g}} \left[ \left\{ 3r^4 \left( -\frac{\Lambda L^2}{6} - \Psi + \hat{\mu}_2 \Psi^2 + \hat{\mu}_3 \Psi^3 + \hat{\mu}_4 \Psi^4 + \hat{\mu}_5 \Psi^5 \right) \right\}' + \frac{q^2 r^3}{2f} (g(rh' + zh)^2 + m^2 L^2 h^2) \right], \quad (10)$$

where

$$\Psi(r) = g(r) - \frac{kL^2}{r^2}, \quad (11)$$

and a prime indicates differentiation with respect to  $r$ .

Independent variation of the reduced action with respect to  $g(r)$ ,  $f(r)$ , and  $h(r)$  produces three ordinary differential equations. The first field equation reads

$$\begin{aligned} 0 = & \Lambda L^2 r^{10} + 3\hat{\mu}_5(3-5z)g^5 r^{10} + 6\hat{\mu}_4(1-2z)g^4 r^{10} + (3z+3)r^{10}g - 6z\hat{\mu}_2 r^{10}g^2 \\ & + 15\hat{\mu}_5 L^2 k(4z-3)g^4 r^8 + 12\hat{\mu}_4 L^2 k(3z-2)g^3 r^8 + 6z\hat{\mu}_2 r^8 L^2 k g - 3r^8 L^2 k \\ & - (9z-3)\hat{\mu}_3 r^{10}g^3 + (18z-9)\hat{\mu}_3 r^8 L^2 k g^2 + 90\hat{\mu}_5 L^4 k^2(1-z)g^3 r^6 \\ & + 36\hat{\mu}_4 L^4 k^2(1-z)r^6 - (9z-9)\hat{\mu}_3 L^4 k^2 r^6 g + 30\hat{\mu}_5 L^6 k(2z-3)g^2 r^4 \\ & + 12\hat{\mu}_4 L^6 k(z-2)gr^4 - 3\hat{\mu}_3 L^6 k^3 r^4 + 6\hat{\mu}_4 L^8 k^2 r^2 + 15(3-z)\hat{\mu}_5 L^8 k^2 gr^2 - 9\hat{\mu}_5 L^{10} k \\ & + g(\ln f)' \left[ \frac{3}{2}r^{11} - 3\hat{\mu}_2 r^{11}g + 3\hat{\mu}_2 r^9 L^2 k - \frac{15}{2}\hat{\mu}_5 r^{11}g^4 - \frac{9}{2}\hat{\mu}_3 r^{11}g^2 \right. \\ & \quad + 30\hat{\mu}_5 r^9 g^3 L^2 k + 9\hat{\mu}_3 r^9 g L^2 k - \frac{9}{2}\hat{\mu}_3 r^7 L^4 k^2 - 6\hat{\mu}_4 g^3 r^{11} \\ & \quad + 18\hat{\mu}_4 L^2 k g^2 r^9 - 45\hat{\mu}_5 L^4 k^2 g^2 r^7 - 18\hat{\mu}_4 L^4 k^2 gr^7 + 6\hat{\mu}_4 r^5 L^6 k \\ & \quad \left. + 30\hat{\mu}_5 r^5 L^6 k g - \frac{15}{2}\hat{\mu}_5 L^8 k^2 r^3 \right] - \frac{q^2 r^{10}}{4f} [g(rh' + zh)^2 - m^2 L^2 h^2]. \end{aligned} \quad (12)$$

The second equation reduces to a simple total derivative,

$$\left[ 3r^4 \left( -\frac{\Lambda L^2}{6} - \Psi + \hat{\mu}_2 \Psi^2 + \hat{\mu}_3 \Psi^3 + \hat{\mu}_4 \Psi^4 + \hat{\mu}_5 \Psi^5 \right) \right]' = \frac{q^2 r^3}{2f} [g(rh' + zh)^2 + m^2 L^2 h^2]. \quad (13)$$

Lastly, variation of the vector-field profile produces

$$2r^2 h'' - r[(\ln f)' - (\ln g)'](rh' + zh) + 2(z+4)rh' + 6zh = 2m^2 L^2 \frac{h}{g}. \quad (14)$$

The radial evolution of the functions  $f(r)$ ,  $g(r)$ , and  $h(r)$  is governed by Eqs. (12)–(14). In the sections that follow, we first use these equations to determine Lifshitz backgrounds and then to construct black-hole solutions that satisfy the near-horizon and asymptotic boundary conditions.

### 3 The conserved quantity

The reduced field equations admit a first integral that is conserved along the radial direction. This conserved quantity is valuable both as a check on the numerical solutions and as a concise means of connecting the near-horizon data with the asymptotic behavior of the fields. Even though the black-hole solutions we construct later are in five dimensions, the conserved charge admits a simple expression in  $(n+1)$  dimensions.

Following the approach employed in the lower-order quasi-topological Lifshitz cases, we define

$$\begin{aligned} F(r) &= \frac{1}{2} \ln f(r) + z \ln \frac{r}{L}, \\ G(r) &= -\frac{1}{2} \ln g(r) - \ln \frac{r}{L}, \\ R(r) &= \ln \frac{r}{L}, \\ H(r) &= \ln h(r) + z \ln \frac{r}{L}. \end{aligned} \tag{15}$$

With these definitions the metric can be expressed as

$$ds^2 = -e^{2F(r)} dt^2 + e^{2G(r)} dr^2 + L^2 e^{2R(r)} d\Omega_{k,n-1}^2, \tag{16}$$

where  $d\Omega_{k,n-1}^2$  denotes the metric of the constant-curvature  $(n-1)$ -dimensional hypersurface.

Upon inserting the metric (16) into the action and integrating by parts, the system reduces to a one-dimensional Lagrangian,

$$\mathcal{L}_{1D} = \mathcal{L}_{1g} + \mathcal{L}_{1m}. \tag{17}$$

The gravitational part of the Lagrangian reads

$$\begin{aligned} \mathcal{L}_{1g} = (n-1) & \left[ -\frac{2\Lambda}{n-1} e^{2G} + (2F'R' + (n-1)R'^2) \right. \\ & - \frac{\hat{\mu}_2 L^2}{3} (4F'R'^3 + (n-4)R'^4) e^{-2G} \\ & - \frac{\hat{\mu}_3 L^4}{5} (6F'R'^5 + (n-6)R'^6) e^{-4G} \\ & - \frac{\hat{\mu}_4 L^6}{7} (8F'R'^7 + (n-8)R'^8) e^{-6G} \\ & \left. - \frac{\hat{\mu}_5 L^8}{9} (10F'R'^9 + (n-10)R'^{10}) e^{-8G} \right] e^{F-G+(n-1)R}, \end{aligned} \tag{18}$$

and the matter contribution is

$$\mathcal{L}_{1m} = \frac{1}{2} q^2 (m^2 + H'^2 e^{-2G}) e^{-F+G+(n-1)R+2H}. \tag{19}$$

Throughout, a prime denotes differentiation with respect to  $r$ .

The equations derived from  $\mathcal{L}_{1D}$  can be arranged into total derivatives. In particular, one finds

$$\mathcal{L}_{1g} - \mathcal{L}_{1m} = \left\{ 2(n-1)e^{F-G+(n-2)R} \left[ R' - \frac{2}{3}\hat{\mu}_2 L^2 e^{-2G} R'^3 - \frac{3}{5}\hat{\mu}_3 L^4 e^{-4G} R'^5 - \frac{4}{7}\hat{\mu}_4 L^6 e^{-6G} R'^7 - \frac{5}{9}\hat{\mu}_5 L^8 e^{-8G} R'^9 \right] \right\}', \quad (20)$$

and

$$\begin{aligned} \mathcal{L}_{1g} + \mathcal{L}_{1m} = & \left\{ e^{F-G+(n-1)R} \left[ 2F' + 2(n-1)R' \right. \right. \\ & - \frac{\hat{\mu}_2 L^2}{3} e^{-2G} (12F' R'^2 + 4(n-4)R'^3) \\ & - \frac{\hat{\mu}_3 L^4}{5} e^{-4G} (30F' R'^4 + 6(n-6)R'^5) \\ & - \frac{\hat{\mu}_4 L^6}{7} e^{-6G} (56F' R'^6 + 8(n-8)R'^7) \\ & \left. \left. - \frac{\hat{\mu}_5 L^8}{9} e^{-8G} (90F' R'^8 + 10(n-10)R'^9) \right] \right\}'. \end{aligned} \quad (21)$$

From the matter sector one obtains

$$2\mathcal{L}_{1m} = \{q^2 H' e^{-F-G+(n-1)R+2H}\}'. \quad (22)$$

Together, Eqs. (20) – (22) combine into a total derivative, yielding a radially conserved quantity. This yields

$$\begin{aligned} \mathcal{C}_0 = & 2(F' - R') \left( 1 - 2\hat{\mu}_2 L^2 R'^2 e^{-2G} - 3\hat{\mu}_3 L^4 R'^4 e^{-4G} - 4\hat{\mu}_4 L^6 R'^6 e^{-6G} - 5\hat{\mu}_5 L^8 R'^8 e^{-8G} \right) e^{F-G+(n-1)R} \\ & - q^2 H' e^{-F-G+(n-1)R+2H}. \end{aligned} \quad (23)$$

When expressed in terms of the original functions  $f(r)$ ,  $g(r)$ , and  $h(r)$ , the conserved charge takes the form

$$\begin{aligned} \mathcal{C}_0 = & \left[ \left( 1 - 2\hat{\mu}_2 g - 3\hat{\mu}_3 g^2 - 4\hat{\mu}_4 g^3 - 5\hat{\mu}_5 g^4 \right) (r f' + 2(z-1)f) \right. \\ & \left. - q^2 (zh + r h') h \right] \frac{r^{z+n-1}}{L^{z+1}} \left( \frac{f}{g} \right)^{1/2}. \end{aligned} \quad (24)$$

This result extends the conserved quantities known from cubic and quartic quasi-topological Lifshitz gravity. It also offers a practical consistency check for the numerical solutions that follow: once the equations of motion are satisfied,  $\mathcal{C}_0$  cannot depend on  $r$ .

In the AdS case  $z = 1$ , with the vector field turned off and  $f(r) = g(r)$ , Eq. (24) simplifies to

$$\mathcal{C}_0 = \frac{r^{n+1}}{L^2} \left( f - \hat{\mu}_2 f^2 - \hat{\mu}_3 f^3 - \hat{\mu}_4 f^4 - \hat{\mu}_5 f^5 \right)'. \quad (25)$$

This represents the quintic extension of the first integral known from lower-order quasi-topological black holes. In asymptotically AdS backgrounds it is proportional to the mass parameter of the black hole.

## 4 Lifshitz solutions

In this section we work out the conditions that allow Lifshitz backgrounds in the quintic quasi-topological theory. We treat the matter-free case first and then turn on the massive vector field. The analysis proceeds directly from the reduced equations derived in Sec. 2.

### 4.1 Matter-free solutions

We start by turning off the vector field,

$$h(r) = 0. \quad (26)$$

A planar Lifshitz background in five dimensions is given by

$$ds^2 = -\frac{r^{2z}}{L^{2z}} dt^2 + \frac{L^2 dr^2}{r^2} + r^2 \sum_{i=1}^3 d\theta_i^2, \quad (27)$$

which amounts to setting  $k = 0$  and  $f(r) = g(r) = 1$ . Plugging this metric into the field equations yields the algebraic conditions

$$\Lambda = -\frac{2}{L^2} (2\hat{\mu}_5 + \hat{\mu}_4 + 2 - \hat{\mu}_2), \quad (28)$$

and

$$\hat{\mu}_3 = -\frac{1}{3} (5\hat{\mu}_5 + 4\hat{\mu}_4 - 1 + 2\hat{\mu}_2). \quad (29)$$

Hence, once  $\hat{\mu}_2$ ,  $\hat{\mu}_4$ , and  $\hat{\mu}_5$  are chosen, the cosmological constant and the cubic coupling are determined by the requirement that the theory possess a Lifshitz vacuum of the form (27). These conditions hold for any value of the dynamical exponent  $z$ .

With Eqs. (28) and (29), the equation following from (13) can be integrated once, leading to

$$\begin{aligned} 2 - \hat{\mu}_2 + \hat{\mu}_4 + 2\hat{\mu}_5 - 3\Psi + 3\hat{\mu}_2\Psi^2 + (1 - 2\hat{\mu}_2 - 4\hat{\mu}_4 - 5\hat{\mu}_5)\Psi^3 \\ + 3\hat{\mu}_4\Psi^4 + 3\hat{\mu}_5\Psi^5 = \frac{C}{r^4}, \end{aligned} \quad (30)$$

where  $C$  is an integration constant and

$$\Psi(r) = g(r) - \frac{kL^2}{r^2}. \quad (31)$$

When  $C = 0$ , the root

$$\Psi = 1 \quad (32)$$

satisfies Eq. (30). Hence,

$$g(r) = 1 + \frac{kL^2}{r^2}. \quad (33)$$

In the hyperbolic case  $k = -1$ , setting  $f(r) = g(r)$  yields

$$f(r) = g(r) = 1 - \frac{L^2}{r^2}, \quad (34)$$

and the metric takes the form

$$ds^2 = -\frac{r^{2z}}{L^{2z}} \left(1 - \frac{L^2}{r^2}\right) dt^2 + \frac{L^2 dr^2}{r^2 \left(1 - \frac{L^2}{r^2}\right)} + r^2 d\Omega_{-1,3}^2. \quad (35)$$

This geometry possesses a horizon at  $r = L$  and constitutes the quintic generalization of the exact hyperbolic branch found in the lower-order quasi-topological Lifshitz theories.

For  $C \neq 0$ , Eq. (30) determines  $g(r)$  implicitly via a quintic algebraic equation. Unlike the cubic and quartic cases, a closed-form expression for the physical branch is not expected in general. Moreover, the mere existence of a real root of the polynomial does not guarantee the correct asymptotic behavior of the full metric; the remaining field equation must also be obeyed by  $f(r)$ . Accordingly, the nontrivial black-hole solutions we are interested in will be built numerically later on.

## 4.2 Solutions supported by a massive vector field

Now we include the massive vector field (9). For the exact Lifshitz background,

$$f(r) = g(r) = h(r) = 1, \quad (36)$$

the field equations fix algebraic relations between the vector charge, the vector mass, the cosmological constant, and the quasi-topological couplings. One finds

$$q^2 = \frac{2(z-1)(1-2\hat{\mu}_2-3\hat{\mu}_3-4\hat{\mu}_4-5\hat{\mu}_5)}{z}, \quad (37)$$

$$m^2 = \frac{(n-1)z}{L^2}, \quad (38)$$

and

$$\Lambda = -\frac{1}{2L^2} \left[ (1-2\hat{\mu}_2-3\hat{\mu}_3-4\hat{\mu}_4-5\hat{\mu}_5)((z-1)^2 + n(z-2) + n^2) + n(n-1)(\hat{\mu}_2 + 2\hat{\mu}_3 + 3\hat{\mu}_4 + 4\hat{\mu}_5) \right]. \quad (39)$$

In five dimensions we set  $n = 4$ , which gives

$$m^2 = \frac{3z}{L^2}. \quad (40)$$

The requirement that the vector charge be real imposes  $q^2 > 0$ . For  $z > 1$  this implies

$$1 - 2\hat{\mu}_2 - 3\hat{\mu}_3 - 4\hat{\mu}_4 - 5\hat{\mu}_5 > 0, \quad (41)$$

or equivalently

$$\hat{\mu}_2 < \frac{1}{2}(1 - 3\hat{\mu}_3 - 4\hat{\mu}_4 - 5\hat{\mu}_5). \quad (42)$$

For  $z = 1$ , Eq. (37) yields  $q = 0$ , and the Lifshitz background reduces to the asymptotically AdS branch.

With the further restriction to the conventional branch  $\hat{\mu}_2 \geq 0$ , the allowed range of  $\Lambda$  dictated by (42) becomes

$$\begin{aligned}
& -\frac{1}{2L^2} \left[ (1 - 3\hat{\mu}_3 - 4\hat{\mu}_4 - 5\hat{\mu}_5) ((z-1)^2 + n(z-2) + n^2) \right. \\
& \quad \left. + n(n-1)(2\hat{\mu}_3 + 3\hat{\mu}_4 + 4\hat{\mu}_5) \right] \leq \Lambda \\
& \leq -\frac{n(n-1)}{4L^2} (1 + \hat{\mu}_3 + 2\hat{\mu}_4 + 3\hat{\mu}_5). \tag{43}
\end{aligned}$$

This range does not constitute an extra assumption; it merely follows from Eq. (39) when the condition  $q^2 > 0$  is imposed on the standard branch.

For  $z > 1$ , the algebraic relations given above ensure the existence of Lifshitz backgrounds supported by the massive vector field. The associated black-hole geometries call for non-trivial radial profiles  $f(r)$ ,  $g(r)$ , and  $h(r)$  that approach unity at infinity and display the proper vanishing behavior at the horizon. Closed-form solutions are not available for the generic quintic theory. Consequently, in the following section we build them using near-horizon series expansions together with a numerical shooting method.

## 5 Asymptotically Lifshitz black holes

We now construct black-hole solutions with nontrivial radial dependence. At the horizon the metric functions must go to zero so that the geometry possesses a regular nonextremal Killing horizon, while at infinity they must recover the Lifshitz background. Hence, for asymptotically Lifshitz black holes we require

$$\lim_{r \rightarrow \infty} f(r) = 1, \quad \lim_{r \rightarrow \infty} g(r) = 1, \quad \lim_{r \rightarrow \infty} h(r) = 1, \tag{44}$$

along with the condition that both  $f(r)$  and  $g(r)$  vanish linearly at the horizon  $r = r_0$ . At  $z = 1$ , Eq. (37) forces the vector charge to zero, which means the gauge field drops out of the gravitational sector entirely. The boundary condition on  $h(r)$  therefore matters only for the matter-supported branches, i.e. when  $z > 1$ .

Because the generic quintic theory does not admit closed-form solutions, the black holes are obtained numerically. The procedure follows the same logic employed in the cubic and quartic quasi-topological Lifshitz cases: we first construct a regular near-horizon expansion, use it to supply initial data just outside the horizon, and then integrate the equations outward, adjusting the shooting parameters until the Lifshitz boundary conditions (44) are met.

### 5.1 Near-horizon expansion

Close to the event horizon at  $r = r_0$  we take the metric functions to vanish linearly. This motivates the expansions

$$\begin{aligned}
f(r) &= f_1 \left[ (r - r_0) + f_2(r - r_0)^2 + f_3(r - r_0)^3 + \cdots \right], \\
g(r) &= g_1(r - r_0) + g_2(r - r_0)^2 + g_3(r - r_0)^3 + \cdots, \\
h(r) &= f_1^{1/2} \left[ h_0 + h_1(r - r_0) + h_2(r - r_0)^2 + h_3(r - r_0)^3 + \cdots \right].
\end{aligned} \tag{45}$$

Inserting these expansions into the field equations gives

$$h_0 = 0. \quad (46)$$

Once  $h_0$  is known,  $g_1$  follows algebraically from  $r_0$ ,  $h_1$ ,  $z$ ,  $k$ ,  $L$ , and the quasi-topological couplings. It is convenient to express  $g_1$  in the compact form

$$g_1 = \frac{z}{r_0^3} \frac{\mathcal{N}_g}{\mathcal{D}_g}, \quad (47)$$

with

$$\begin{aligned} \mathcal{N}_g = & \hat{\mu}_4 \left[ (4z^2 + 8z)r_0^{10} + 12L^8 k^4 r_0^2 \right] + \hat{\mu}_3 \left[ (3z^2 + 6z + 3)r_0^{10} - 6L^6 k^3 r_0^4 \right] \\ & + \hat{\mu}_2 \left[ (2z^2 + 4z + 6)r_0^{10} \right] + \hat{\mu}_5 \left[ (5z^2 + 10z - 3)r_0^{10} - 18L^{10} k^5 \right] \\ & - (z^2 + 2z + 9)r_0^{10} - 6L^2 k r_0^8, \end{aligned} \quad (48)$$

and

$$\begin{aligned} \mathcal{D}_g = & \hat{\mu}_4 \left[ (4h_1^2 z - 4h_1^2)r_0^9 - 12zL^6 k^3 r_0^2 \right] + \hat{\mu}_3 \left[ (3h_1^2 z - 3h_1^2)r_0^9 + 9zL^4 k^2 r_0^4 \right] \\ & + \hat{\mu}_2 \left[ (2h_1^2 z - 2h_1^2)r_0^9 - 6zL^2 k r_0^6 \right] + \hat{\mu}_5 \left[ (5h_1^2 z - 5h_1^2)r_0^9 + 15zL^8 k^4 \right] \\ & + (-h_1^2 z + h_1^2)r_0^9 - 3zr_0^8. \end{aligned} \quad (49)$$

All remaining coefficients in the expansions follow recursively. For instance,  $h_2$ ,  $f_2$ , and  $g_2$  are completely fixed once  $f_1$ ,  $h_1$ , and the theory parameters are specified. The first non-trivial gauge-field coefficient is given in Appendix B; all higher coefficients are obtained recursively by inserting the near-horizon series into the field equations order by order. During the numerical integration, we use these recursion relations to build the initial data at  $r = r_0 + \epsilon$ . What matters for the numerical construction is that  $f_1$  and  $h_1$  remain free and serve as the shooting parameters. These parameters are tuned so that the solution approaches the Lifshitz form at large  $r$ .

## 5.2 Asymptotic behavior

Far from the horizon the solutions must relax to the Lifshitz background. We thus expand

$$f(r) = 1 + \delta f(r), \quad g(r) = 1 + \delta g(r), \quad h(r) = 1 + \delta h(r), \quad (50)$$

and require that all three perturbations vanish as  $r \rightarrow \infty$ :

$$\lim_{r \rightarrow \infty} \delta f(r) = 0, \quad \lim_{r \rightarrow \infty} \delta g(r) = 0, \quad \lim_{r \rightarrow \infty} \delta h(r) = 0. \quad (51)$$

The linearized field equations dictate which falloff modes are permitted. For the numerical construction carried out here, the explicit falloff exponents are not required analytically; rather, the shooting parameters are chosen so that the integrated solution meets the Lifshitz boundary conditions to within the prescribed numerical tolerance. In practice, the large- $r$  behavior is used only to set the boundary condition and not as a standalone analytic solution. After integrating outward from  $r = r_0 + \epsilon$ , we adjust the free near-horizon parameters so that the numerical solution satisfies Eq. (44). This constitutes the shooting procedure employed in what follows.

### 5.3 First-order system for numerical integration

For numerical work we recast the field equations as a system of first-order ordinary differential equations. We define

$$j(r) = h'(r), \quad (52)$$

so that

$$h'(r) = j(r). \quad (53)$$

From the remaining equations one then extracts explicit expressions for  $f'(r)$ ,  $g'(r)$ , and  $j'(r)$ . To write the system in a compact form we introduce

$$X(r) = rj(r) + zh(r), \quad (54)$$

together with

$$\Psi(r) = g(r) - \frac{kL^2}{r^2}. \quad (55)$$

We further define

$$\mathcal{P}(\Psi) = -\frac{\Lambda L^2}{6} - \Psi + \hat{\mu}_2 \Psi^2 + \hat{\mu}_3 \Psi^3 + \hat{\mu}_4 \Psi^4 + \hat{\mu}_5 \Psi^5. \quad (56)$$

With these definitions the first gravitational equation takes the form

$$\mathcal{A}(r, g) + g(r) \frac{f'(r)}{f(r)} \mathcal{B}(r, g) = \frac{q^2 r^{10}}{4f(r)} \left[ g(r) X^2(r) - m^2 L^2 h^2(r) \right], \quad (57)$$

where

$$\begin{aligned} \mathcal{A}(r, g) = & \Lambda L^2 r^{10} + 3\hat{\mu}_5(3-5z)g^5 r^{10} + 6\hat{\mu}_4(1-2z)g^4 r^{10} + (3z+3)r^{10}g - 6z\hat{\mu}_2 r^{10}g^2 \\ & + 15\hat{\mu}_5 L^2 k(4z-3)g^4 r^8 + 12\hat{\mu}_4 L^2 k(3z-2)g^3 r^8 + 6z\hat{\mu}_2 r^8 L^2 k g - 3r^8 L^2 k \\ & - (9z-3)\hat{\mu}_3 r^{10}g^3 + (18z-9)\hat{\mu}_3 r^8 L^2 k g^2 + 90\hat{\mu}_5 L^4 k^2(1-z)g^3 r^6 \\ & + 36\hat{\mu}_4 L^4 k^2(1-z)r^6 - (9z-9)\hat{\mu}_3 L^4 k^2 r^6 g + 30\hat{\mu}_5 L^6 k(2z-3)g^2 r^4 \\ & + 12\hat{\mu}_4 L^6 k(z-2)gr^4 - 3\hat{\mu}_3 L^6 k^3 r^4 + 6\hat{\mu}_4 L^8 k^2 r^2 \\ & + 15(3-z)\hat{\mu}_5 L^8 k^2 gr^2 - 9\hat{\mu}_5 L^{10} k, \end{aligned} \quad (58)$$

and

$$\begin{aligned} \mathcal{B}(r, g) = & \frac{3}{2}r^{11} - 3\hat{\mu}_2 r^{11}g + 3\hat{\mu}_2 r^9 L^2 k - \frac{15}{2}\hat{\mu}_5 r^{11}g^4 - \frac{9}{2}\hat{\mu}_3 r^{11}g^2 \\ & + 30\hat{\mu}_5 r^9 g^3 L^2 k + 9\hat{\mu}_3 r^9 g L^2 k - \frac{9}{2}\hat{\mu}_3 r^7 L^4 k^2 - 6\hat{\mu}_4 g^3 r^{11} \\ & + 18\hat{\mu}_4 L^2 k g^2 r^9 - 45\hat{\mu}_5 L^4 k^2 g^2 r^7 - 18\hat{\mu}_4 L^4 k^2 gr^7 + 6\hat{\mu}_4 r^5 L^6 k \\ & + 30\hat{\mu}_5 r^5 L^6 k g - \frac{15}{2}\hat{\mu}_5 L^8 k^2 r^3. \end{aligned} \quad (59)$$

The second gravitational equation can be expressed as

$$\frac{d}{dr} \left[ 3r^4 \mathcal{P}(\Psi) \right] = \frac{q^2 r^3}{2f(r)} \left[ g(r) X^2(r) + m^2 L^2 h^2(r) \right], \quad (60)$$

while the equation for the vector field takes the form

$$2r^2 j'(r) - r \left( \frac{f'(r)}{f(r)} - \frac{g'(r)}{g(r)} \right) X(r) + 2(z+4)rj(r) + 6zh(r) = \frac{2m^2 L^2 h(r)}{g(r)}. \quad (61)$$

Table 1: Parameter values adopted for the numerical solutions shown in the figures. In the figure panels,  $\mu_i$  stands for the dimensionless coupling  $\hat{\mu}_i$  introduced in the text.

| Figure    | $z$ | $k$ | $r_0$ | $\hat{\mu}_2$ | $\hat{\mu}_3$ | $\hat{\mu}_4$ | $\hat{\mu}_5$ |
|-----------|-----|-----|-------|---------------|---------------|---------------|---------------|
| Fig. 1(a) | 1   | -1  | 1.68  | 0.4           | -0.1          | -0.001        | 0.001         |
| Fig. 1(b) | 1   | 0   | 2.13  | 0.04          | -0.001        | 0.0001        | -0.0001       |
| Fig. 1(c) | 1   | +1  | 2.91  | 0.4           | -0.01         | 0.001         | -0.001        |
| Fig. 2(a) | 2   | -1  | 1.5   | 0.04          | -0.001        | 0.0001        | -0.0001       |
| Fig. 2(b) | 2   | 0   | 3.0   | 0.04          | -0.001        | 0.0001        | -0.0001       |
| Fig. 2(c) | 2   | +1  | 4.0   | 0.04          | -0.001        | 0.0001        | -0.0001       |
| Fig. 3(a) | 2   | -1  | 1.5   | 0.04          | -0.001        | 0.0001        | -0.0001       |
| Fig. 3(b) | 2   | 0   | 2.0   | 0.04          | -0.001        | 0.0001        | -0.0001       |
| Fig. 3(c) | 2   | +1  | 4.0   | 0.04          | -0.001        | 0.0001        | -0.0001       |
| Fig. 4(a) | 2   | -1  | 1.5   | 0.04          | -0.001        | 0.0001        | -0.0001       |
| Fig. 4(b) | 2   | 0   | 3.0   | 0.04          | -0.001        | 0.0001        | -0.0001       |
| Fig. 4(c) | 2   | +1  | 4.0   | 0.04          | -0.001        | 0.0001        | -0.0001       |

The numerical shooting procedure is based on the set of equations (53), (57), (60), and (61). In practice, one solves them algebraically for  $h'(r)$ ,  $f'(r)$ ,  $g'(r)$ , and  $j'(r)$ , which gives four coupled first-order equations governing the variables

$$\{f(r), g(r), h(r), j(r)\}. \quad (62)$$

The integration begins at  $r = r_0 + \epsilon$ , with  $\epsilon$  a small positive number. The initial values of  $f$ ,  $g$ ,  $h$ , and  $j$  are read off from the near-horizon series (45). The shooting parameters are then adjusted until the large-radius behavior matches Eq. (44). In the numerical runs reported here, we integrated outward to a large cutoff radius  $r_{\max}$  and tuned the shooting parameters until the quantities  $|f(r_{\max}) - 1|$ ,  $|g(r_{\max}) - 1|$ , and  $|h(r_{\max}) - 1|$  lay below the numerical tolerance of the solver. The asymptotically Lifshitz black-hole solutions presented in the following section are obtained in this way.

## 6 Numerical solutions

We now turn to the numerical black-hole solutions that follow from the first-order system introduced in Sec. 5.3. Starting the integration just outside the horizon with the help of the near-horizon expansion, we then tune the remaining free parameters so that at large radius the metric functions and the gauge-field profile relax to their Lifshitz form. We treat the two cases  $z = 1$  and  $z = 2$  separately, and for each dynamical exponent we examine all three horizon topologies  $k = -1, 0, +1$ . The parameter choices for every numerical run are listed in the respective figure captions. Throughout the figures, the symbols  $\mu_i$  refer to the dimensionless couplings  $\hat{\mu}_i$  employed in the main text.

We set  $L = 1$  in all numerical examples. Given a choice of  $z$  and the quasi-topological couplings, the Lifshitz conditions (37)–(39) then determine  $q$ ,  $m$ , and  $\Lambda$  uniquely. The

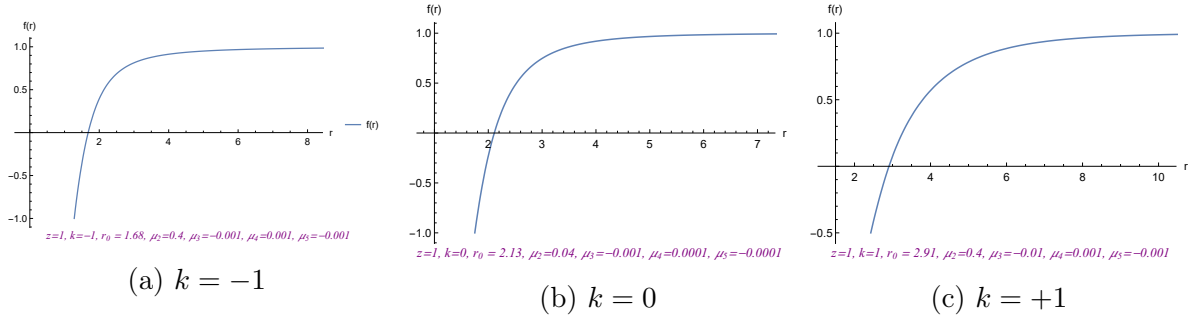


Figure 1: Metric function  $f(r)$  for the  $z = 1$  branch, shown for the three horizon topologies  $k = -1, 0, +1$ . The coupling parameters adopted in each panel are given on the plots. In every case  $f(r)$  vanishes at the horizon and goes to unity at large  $r$ , as required by the asymptotically AdS condition.

independent numerical inputs are therefore the horizon topology, the horizon radius, the curvature couplings, and the shooting parameters that appear in the near-horizon expansion. The coefficients  $f_1$  and  $h_1$  appearing in the near-horizon expansion serve as shooting parameters. For every row listed in Table 1, these two parameters are tuned so that the resulting numerical solution obeys the asymptotically Lifshitz boundary conditions at large radius. If one wishes, the numerical values of  $f_1$  and  $h_1$  can also be listed in an extended version of the table, but the physically essential parameters of the displayed branches are already those collected in Table 1.

The aim of the numerical examples shown here is to establish that quintic quasi-topological Lifshitz black holes exist, and to illustrate their qualitative behavior for a representative set of coupling values. A comprehensive exploration of the full quintic coupling space lies outside the scope of this paper.

We begin with the  $z = 1$  branch. Here the Lifshitz scaling collapses to the relativistic AdS case, and the massive vector charge goes to zero. The  $z = 1$  branch thus offers a useful consistency check on the numerical construction before one proceeds to the genuinely Lifshitz geometries with  $z \neq 1$ . Figure 1 shows the metric function  $f(r)$  for the three horizon topologies. In every case shown,  $f(r)$  passes through zero at the horizon and tends to unity at large  $r$ , consistent with the asymptotically AdS boundary condition. The detailed shape of the radial profile varies with the horizon curvature and the choice of quasi-topological couplings, but the overall behavior matches what is seen in lower-order quasi-topological black holes.

We now move to the genuine Lifshitz case  $z = 2$ . The metric function  $f(r)$  for this branch is displayed in Fig. 2 for the three topologies  $k = -1, 0, +1$ . Relative to the  $z = 1$  solutions, those with  $z = 2$  exhibit a more pronounced radial variation outside the horizon. This pattern agrees with what is observed in the lower-order Lifshitz quasi-topological black holes, where the metric functions can develop a noticeable transient hump before settling to their asymptotic values. How pronounced this feature is depends on both the horizon topology and the horizon radius adopted in the shooting procedure. Among the examples displayed, the hyperbolic case shows the strongest enhancement of  $f(r)$ , whereas the planar and spherical cases produce milder profiles for the same set of quasi-topological couplings.

Figure 3 provides a closer look at the metric sector for  $z = 2$ , with the two functions  $f(r)$  and  $g(r)$  displayed together. Both functions go to zero at the horizon and relax to their Lifshitz asymptotics at large radius, yet they are not forced to match each

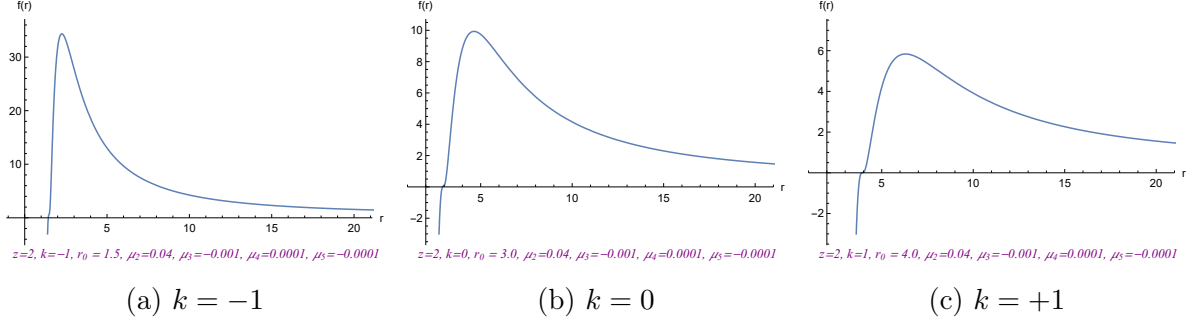


Figure 2: Metric function  $f(r)$  for the  $z = 2$  Lifshitz branch, plotted for the three horizon topologies  $k = -1, 0, +1$ . The coupling values used in each panel are indicated on the respective plots. All solutions vanish at the horizon and approach the Lifshitz value at large  $r$ . Relative to the  $z = 1$  case, the radial profiles display a stronger intermediate variation, a feature expected for Lifshitz black holes.

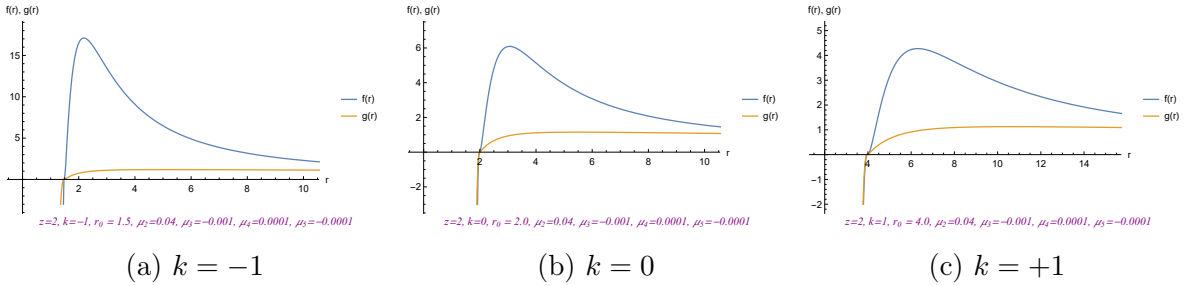


Figure 3: Comparison of  $f(r)$  and  $g(r)$  for the  $z = 2$  Lifshitz branch, shown for  $k = -1, 0, +1$ . The parameter choices are listed on the individual panels. Both functions vanish at the horizon and approach the Lifshitz asymptotics at large radius. Away from the horizon they differ, as expected for a generic Lifshitz black-hole ansatz.

other away from the horizon. This distinction matters for Lifshitz black holes, because the reduced equations do not generically enforce  $f(r) = g(r)$ . The profiles reveal a correlated qualitative behavior between the two functions: whenever  $f(r)$  develops a stronger intermediate peak, the corresponding  $g(r)$  profile follows the same overall pattern. As in the preceding figure, the hyperbolic case exhibits the strongest radial variation, while the planar and spherical cases remain milder for the representative parameters used here.

Figure 4 presents the full numerical solution for the  $z = 2$  branch, showing the three functions  $f(r)$ ,  $g(r)$ , and  $h(r)$  together. Since the Proca field is essential to sustain the Lifshitz asymptotics when  $z \neq 1$ , its radial profile forms an integral part of the black-hole solution and cannot be regarded as an independent probe. In the examples displayed,  $h(r)$  remains regular outside the horizon and approaches its asymptotic limit alongside the metric functions. All three functions exhibit the same qualitative behavior seen in the cubic and quartic quasi-topological Lifshitz cases: after departing from the near-horizon region, the solution can develop an intermediate peak before settling into the Lifshitz background at large radius. This feature is most prominent in the hyperbolic and planar examples presented here, whereas the spherical case displays a comparatively gentler radial profile for the chosen parameter set.

These numerical results demonstrate that the quintic theory supports regular asymptotically Lifshitz black-hole solutions for the representative parameter choices explored

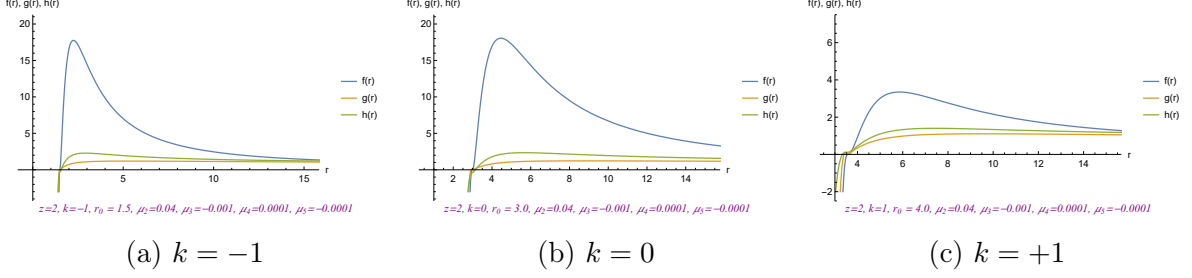


Figure 4: Full set of functions  $f(r)$ ,  $g(r)$ , and  $h(r)$  for the  $z = 2$  Lifshitz branch, plotted for the three horizon topologies  $k = -1, 0, +1$ . The parameter values are indicated within each panel. Together with the metric functions, the gauge-field profile  $h(r)$  approaches its Lifshitz value at large  $r$ .

here. That said, the branches shown are regular and meet the required boundary conditions to within numerical tolerance, but they do not amount to a full classification of all possible solutions in the quintic theory. Other corners of the coupling space could well contain further branches, or they may impose stricter conditions on the shooting parameters. The solutions exhibit the proper near-horizon behavior and asymptote to the Lifshitz background at infinity. Qualitatively, their radial profiles match those encountered in the cubic and quartic quasi-topological theories, while the extra quintic coupling expands the accessible parameter space. In the next section we compute the thermodynamic quantities for these numerical branches and investigate their local thermal behavior.

## 7 Thermodynamics

We now study the thermodynamic properties of the quintic quasi-topological Lifshitz black holes. The temperature follows from the surface gravity at the horizon, and the entropy is obtained via the Wald formula, which is the appropriate prescription for higher-curvature theories. In the thermodynamic discussion that follows, all entropy values are given as densities per unit horizon volume and are expressed in the same gravitational units used for the action.

The entropy is computed using the Iyer–Wald prescription for higher-curvature gravity,

$$S = -2\pi \int_{\Sigma_h} d^{n-1}x \sqrt{\tilde{g}} Y^{abcd} \hat{\epsilon}_{ab} \hat{\epsilon}_{cd}, \quad Y^{abcd} = \frac{\partial \mathcal{L}}{\partial R_{abcd}}, \quad (63)$$

where  $\Sigma_h$  is a spatial cross section of the horizon,  $\tilde{g}$  is the determinant of the induced metric on  $\Sigma_h$ , and  $\hat{\epsilon}_{ab}$  is the horizon binormal.

For the static black-hole solutions under consideration, the entropy density  $\mathcal{S} = S/V_{n-1}$  reads

$$\mathcal{S} = \frac{r_0^{n-1}}{4} \left[ 1 - 2k\hat{\mu}_2 \frac{(n-1)L^2}{(n-3)r_0^2} + 3k^2\hat{\mu}_3 \frac{(n-1)L^4}{(n-5)r_0^4} - 4k\hat{\mu}_4 \frac{(n-1)L^6}{(n-7)r_0^6} + 5k^2\hat{\mu}_5 \frac{(n-1)L^8}{(n-9)r_0^8} \right]. \quad (64)$$

The signs appearing in Eq. (64) are a consequence of the normalization adopted for the dimensionless couplings  $\hat{\mu}_i$  in the reduced action (10). In this convention, one retrieves

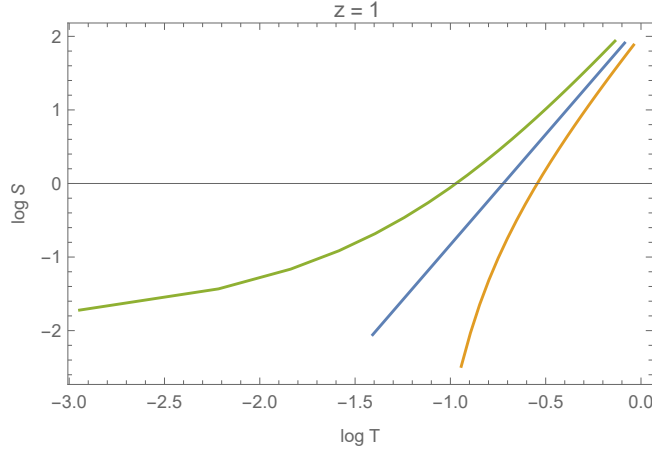


Figure 5: Logarithmic entropy–temperature relation for the  $z = 1$  branch, plotted for all three horizon topologies. The green, blue, and orange curves correspond to  $k = -1$ ,  $k = 0$ , and  $k = +1$ , respectively. The vertical axis shows  $\log \mathcal{S}$  and the horizontal axis  $\log T$ . The positive slope of each curve signals a positive heat capacity for the displayed branches.

the lower-order results simply by taking the appropriate higher-curvature couplings to vanish. Even though the numerical solutions presented in this paper are built in a five-dimensional bulk ( $n = 4$ ), Eq. (64) has been expressed in general  $(n + 1)$ -dimensional form to make transparent the contribution of each curvature order to the Wald entropy. The five-dimensional entropy density appearing in the numerical plots follows from Eq. (64) by taking  $n = 4$ , with the same coupling conventions adopted in the reduced action.

The Hawking temperature follows from the requirement that the Euclidean section be regular at the horizon. This yields

$$T = \left. \frac{r^{z+1} \sqrt{f'(r)g'(r)}}{4\pi L^{z+1}} \right|_{r=r_0}. \quad (65)$$

Given a numerical black-hole solution, one obtains the temperature from Eq. (65) using  $r_0$  and the horizon values of  $f'(r_0)$  and  $g'(r_0)$ , while Eq. (64) supplies the entropy density. From these quantities we then build the logarithmic entropy–temperature plots shown in Figs. 5 and 6.

With the numerical solutions in hand, the temperature is extracted from the surface gravity at the horizon, and the entropy is computed using the Wald formula, which is the correct prescription for higher-curvature theories. To probe the local thermal behavior of the solutions, we analyze the relation between entropy density and temperature using logarithmic variables. We display  $\log \mathcal{S}$  against  $\log T$  for the three horizon topologies  $k = -1, 0, +1$ , covering both the  $z = 1$  and  $z = 2$  cases. Given the heat capacity density

$$C = T \frac{d\mathcal{S}}{dT}, \quad (66)$$

one obtains

$$\frac{d \log \mathcal{S}}{d \log T} = \frac{T}{\mathcal{S}} \frac{d\mathcal{S}}{dT} = \frac{C}{\mathcal{S}}. \quad (67)$$

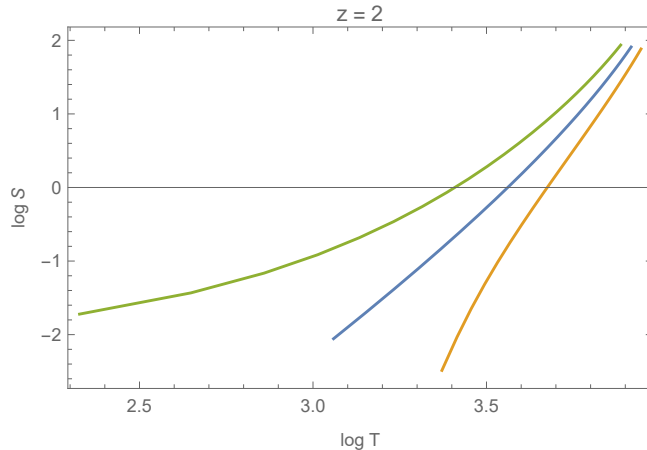


Figure 6: Logarithmic entropy–temperature relation for the  $z = 2$  Lifshitz branch, shown for all three horizon topologies. As before, the green, blue, and orange curves represent  $k = -1$ ,  $k = 0$ , and  $k = +1$ . The axes are  $\log \mathcal{S}$  (vertical) and  $\log T$  (horizontal). The upward slope of the curves indicates that the displayed branches have positive heat capacity.

Hence, for branches with positive entropy density, an upward slope in the  $\log \mathcal{S}$ – $\log T$  plane signals a positive heat capacity. The logarithmic plots therefore serve as a convenient diagnostic for the local thermodynamic stability of the numerical branches studied here.

Figure 5 displays the thermodynamic behavior for  $z = 1$ . At this value the Lifshitz scaling reduces to the relativistic AdS branch and the massive vector charge disappears. The three curves correspond to the horizon topologies  $k = -1, 0, +1$ . Throughout the displayed range,  $\log \mathcal{S}$  grows monotonically with  $\log T$  for all three topologies. Because the entropy density plotted is positive, the upward slope signals a positive heat capacity for each of these branches. The  $z = 1$  black holes shown here are therefore locally stable against thermal fluctuations within the plotted parameter range. The separation between the curves reflects how the horizon topology affects the entropy–temperature relation.

The thermodynamic behavior for  $z = 2$  appears in Fig. 6. As before, the three curves represent  $k = -1, 0, +1$  with the same color coding used in Fig. 5. Across the range shown,  $\log \mathcal{S}$  rises monotonically with  $\log T$  for every topology. With the entropy density remaining positive throughout, the upward slope of these curves signals a positive heat capacity for the  $z = 2$  Lifshitz branches. The solutions shown are therefore locally stable against thermal fluctuations over the parameter range covered by the plots. Relative to the  $z = 1$  case, the curves are displaced toward larger values of  $\log T$ , a shift that reflects the altered temperature scaling tied to the Lifshitz exponent. Even so, the qualitative dependence of the entropy density on temperature remains similar to that seen in the relativistic branch.

## 8 Conclusions

In this paper we have investigated asymptotically Lifshitz black holes in quintic quasi-topological gravity. This theory pushes the quasi-topological construction to fifth order in curvature, introducing an extra higher-curvature coupling while keeping a manage-

able set of reduced field equations for the static constant-curvature ansatz. We worked in a five-dimensional bulk and coupled gravity to a massive Abelian vector field, which is needed to sustain Lifshitz asymptotics when  $z \neq 1$ .

We began with the quintic quasi-topological action, inserted the Lifshitz black-hole ansatz, and worked out the reduced equations that determine the metric functions  $f(r)$ ,  $g(r)$  and the gauge-field profile  $h(r)$ . A radially conserved charge for the one-dimensional reduced system was also constructed. When  $z = 1$ , this charge collapses to the familiar polynomial involving all quasi-topological couplings through quintic order.

We next examined the requirements for Lifshitz backgrounds to exist in the theory. Without matter, the existence of an exact Lifshitz background imposes algebraic relations linking the cosmological constant to the curvature couplings. When the massive vector field is switched on, the charge, the mass, and the cosmological constant become fixed by the dynamical exponent and the quasi-topological couplings. The condition that the vector charge be real then places a corresponding constraint on the admissible region of coupling space.

Given that closed-form black-hole solutions are absent in the generic quintic theory, we turned to numerical construction. Regular initial data were generated just outside the horizon from the near-horizon expansion, and the first-order system was integrated outward while enforcing the asymptotically Lifshitz boundary conditions. We produced numerical solutions for  $z = 1$  and  $z = 2$ , covering the three horizon topologies  $k = -1, 0, +1$ . The metric functions and gauge-field profiles share the overall structure familiar from the cubic and quartic quasi-topological Lifshitz cases: the fields vanish correctly at the horizon, develop radial profiles that depend on the topology and the chosen parameters, and relax to the Lifshitz background at large radius.

We further investigated the thermodynamics of the numerical black holes. The entropy density was evaluated via the Wald prescription appropriate for higher-curvature theories, and the temperature was extracted from the requirement that the Euclidean section be regular at the horizon. For the representative quintic couplings chosen in this study, the logarithmic entropy–temperature plots exhibit a positive slope for all three horizon topologies within the displayed ranges, for both  $z = 1$  and  $z = 2$ . Because these plots use  $\log \mathcal{S}$  and  $\log T$  as axes, and the branches shown carry positive entropy density, a positive slope translates into a positive heat capacity. The displayed numerical branches are therefore locally stable against thermal fluctuations.

These results suggest that the quintic quasi-topological interaction furnishes a consistent higher-order extension of the cubic and quartic Lifshitz black holes studied previously. It expands the coupling space while retaining, for the representative examples examined here, the qualitative structure found in the lower-order solutions.

A more exhaustive exploration of the full quintic parameter space, the role of the shooting parameters, global thermodynamic quantities such as the free energy, and potential holographic restrictions — for instance, causality, positivity of energy flux, or quasinormal-mode stability — would all be valuable avenues for future investigation.

## A Curvature invariants and coupling conventions

This appendix collects the curvature-invariant basis and the coupling conventions adopted in the main body of the paper. The gravitational action includes the Einstein-Hilbert term, the Gauss-Bonnet density, and quasi-topological curvature terms through fifth

order. The first two densities are

$$\mathcal{L}_1 = R, \quad \mathcal{L}_2 = R_{abcd}R^{abcd} - 4R_{ab}R^{ab} + R^2. \quad (68)$$

The cubic quasi-topological density is written as

$$\begin{aligned} \mathcal{L}_3 = & R_a{}^c{}_b{}^d R_c{}^e{}_d{}^f R_e{}^a{}_f{}^b + \frac{1}{(2n-1)(n-3)} \left[ \frac{3(3n-5)}{8} R_{abcd}R^{abcd}R \right. \\ & - 3(n-1)R_{abcd}R^{abc}{}_e R^{de} + 3(n+1)R_{abcd}R^{ac}R^{bd} + 6(n-1)R_a{}^b R_b{}^c R_c{}^a \\ & \left. - \frac{3(3n-1)}{2} R_a{}^b R_b{}^a R + \frac{3(n+1)}{8} R^3 \right]. \end{aligned} \quad (69)$$

The quartic quasi-topological density is expressed as

$$\begin{aligned} \mathcal{L}_4 = & b_1 R_{abcd}R^{cdef}R^{hg}{}_{ef}R_{hg}{}^{ab} + b_2 R_{abcd}R^{abcd}R_{ef}R^{ef} + b_3 R R_{ab}R^{ac}R_c{}^b \\ & + b_4 (R_{abcd}R^{abcd})^2 + b_5 R_{ab}R^{ac}R_{cd}R^{db} + b_6 R R_{abcd}R^{ac}R^{db} \\ & + b_7 R_{abcd}R^{ac}R^{be}R^d{}_e + b_8 R_{abcd}R^{acef}R^b{}_e R^d{}_f + b_9 R_{abcd}R^{ac}R_{ef}R^{bedf} \\ & + b_{10} R^4 + b_{11} R^2 R_{abcd}R^{abcd} + b_{12} R^2 R_{ab}R^{ab} \\ & + b_{13} R_{abcd}R^{abef}R_{ef}{}^c{}_g R^{dg} + b_{14} R_{abcd}R^{aecf}R_{gehf}R^{gbhd}. \end{aligned} \quad (70)$$

The quintic quasi-topological density is written in the invariant basis

$$\begin{aligned} \mathcal{L}_5 = & c_1 R R_b{}^a R_c{}^b R_d{}^c R_a{}^d + c_2 R R_b{}^a R_a{}^b R_{ef}{}^{cd} R_{cd}{}^{ef} + c_3 R R_c{}^a R_d{}^b R_{ef}{}^{cd} R_{ab}{}^{ef} \\ & + c_4 R_b{}^a R_a{}^b R_d{}^c R_e{}^d R_c{}^e + c_5 R_b{}^a R_c{}^b R_a{}^c R_{fg}{}^{de} R_{de}{}^{fg} \\ & + c_6 R_b{}^a R_d{}^b R_f{}^c R_{ag}{}^{de} R_{ce}{}^{fg} + c_7 R_b{}^a R_d{}^b R_f{}^c R_{cg}{}^{de} R_{ae}{}^{fg} \\ & + c_8 R_b{}^a R_c{}^b R_{ae}{}^{cd} R_{gh}{}^{ef} R_{df}{}^{gh} + c_9 R_b{}^a R_c{}^b R_{ef}{}^{cd} R_{gh}{}^{ef} R_{ad}{}^{gh} \\ & + c_{10} R_b{}^a R_c{}^b R_{eg}{}^{cd} R_{ah}{}^{ef} R_{df}{}^{gh} + c_{11} R_c{}^a R_d{}^b R_{ab}{}^{cd} R_{gh}{}^{ef} R_{ef}{}^{gh} \\ & + c_{12} R_c{}^a R_d{}^b R_{ae}{}^{cd} R_{gh}{}^{ef} R_{bf}{}^{gh} + c_{13} R_c{}^a R_d{}^b R_{ef}{}^{cd} R_{gh}{}^{ef} R_{ab}{}^{gh} \\ & + c_{14} R_c{}^a R_d{}^b R_{eg}{}^{cd} R_{ah}{}^{ef} R_{bf}{}^{gh} + c_{15} R_c{}^a R_e{}^b R_{af}{}^{cd} R_{gh}{}^{ef} R_{bd}{}^{gh} \\ & + c_{16} R_b{}^a R_{ad}{}^{bc} R_{fh}{}^{de} R_{ci}{}^{fg} R_{eg}{}^{hi} + c_{17} R_b{}^a R_{de}{}^{bc} R_{cf}{}^{de} R_{hi}{}^{fg} R_{ag}{}^{hi} \\ & + c_{18} R_b{}^a R_{df}{}^{bc} R_{ac}{}^{de} R_{hi}{}^{fg} R_{eg}{}^{hi} + c_{19} R_b{}^a R_{df}{}^{bc} R_{ah}{}^{de} R_{ei}{}^{fg} R_{cg}{}^{hi} \\ & + c_{20} R_b{}^a R_{df}{}^{bc} R_{gh}{}^{de} R_{ei}{}^{fg} R_{ac}{}^{hi} + c_{21} R_{cd}{}^{ab} R_{eg}{}^{cd} R_{ai}{}^{ef} R_{fj}{}^{gh} R_{bh}{}^{ij} \\ & + c_{22} R_{ce}{}^{ab} R_{af}{}^{cd} R_{gi}{}^{ef} R_{bj}{}^{gh} R_{dh}{}^{ij} + c_{23} R_{ce}{}^{ab} R_{ag}{}^{cd} R_{bi}{}^{ef} R_{fj}{}^{gh} R_{dh}{}^{ij} \\ & + c_{24} R_{ce}{}^{ab} R_{fg}{}^{cd} R_{hi}{}^{ef} R_{aj}{}^{gh} R_{bd}{}^{ij}. \end{aligned} \quad (71)$$

The coefficients  $b_i$  and  $c_i$  are fixed by demanding that, for the static constant-curvature ansatz, the reduced equations take the quasi-topological form employed in the main text. For the quartic density we adopt the standard coefficients of quartic quasi-topological gravity, while for the quintic density we use the representative quintic choice given in Ref. [20]. With these conventions, inserting the Lifshitz ansatz into the action produces the reduced polynomial

$$-\Psi + \hat{\mu}_2 \Psi^2 + \hat{\mu}_3 \Psi^3 + \hat{\mu}_4 \Psi^4 + \hat{\mu}_5 \Psi^5, \quad (72)$$

in which

$$\Psi(r) = g(r) - \frac{kL^2}{r^2}. \quad (73)$$

The dimensionless couplings  $\hat{\mu}_i$  that appear in the reduced equations are those used throughout the main text. The aim of this appendix is simply to state the invariant basis and the conventions behind the reduced equations; the full coefficient tables for the quartic and quintic densities are rather lengthy and are taken, respectively, from the quartic and quintic quasi-topological constructions of Refs. [14, 20].

## B Near-horizon coefficients and numerical initial data

This appendix collects the supplementary near-horizon data needed to start the numerical integration. The near-horizon expansions and the condition  $h_0 = 0$  were already presented in Eqs. (45) and (46). The coefficient  $g_1$  is determined by Eq. (47), whereas  $f_1$  and  $h_1$  are left free as shooting parameters.

The first non-trivial coefficient in the gauge-field expansion takes the form

$$h_2 = -\frac{h_1}{2r_0^2 g_1} \frac{\mathcal{N}_h}{\mathcal{D}_h}, \quad (74)$$

with

$$\begin{aligned} \mathcal{N}_h = & \left[ \left( [1 - 2\hat{\mu}_2 - 3\hat{\mu}_3 - 4\hat{\mu}_4 - 5\hat{\mu}_5] h_1^2 - 2g_1 \right) z \right. \\ & \left. + (-1 + 2\hat{\mu}_2 + 3\hat{\mu}_3 + 4\hat{\mu}_4 + 5\hat{\mu}_5) h_1^2 - 3g_1 \right] r_0^9 \\ & + 3zr_0^8 - (4z + 6)\hat{\mu}_2 L^2 k g_1 r_0^7 + 6z\hat{\mu}_2 L^2 k r_0^6 \\ & + (9 + 6z)\hat{\mu}_3 L^4 k^2 g_1 r_0^5 - 9z\hat{\mu}_3 L^4 k^2 r_0^4 \\ & - (8z + 12)\hat{\mu}_4 L^6 k^3 g_1 r_0^3 + 12z\hat{\mu}_4 L^6 k^3 r_0^2 \\ & + (10z + 15)\hat{\mu}_5 L^8 k^4 g_1 r_0 - 15z\hat{\mu}_5 L^8 k^4, \end{aligned} \quad (75)$$

and

$$\mathcal{D}_h = -r_0^8 - 2\hat{\mu}_2 L^2 k r_0^6 + 3\hat{\mu}_3 L^4 k^2 r_0^4 - 4\hat{\mu}_4 L^6 k^3 r_0^2 + 5\hat{\mu}_5 L^8 k^4. \quad (76)$$

The same approach yields  $f_2$ ,  $g_2$ , and the coefficients at all higher orders. At each power of  $(r - r_0)$ , the field equations supply algebraic conditions that involve the unknown expansion coefficients. With the inputs  $r_0$ ,  $k$ ,  $z$ ,  $L$ , the couplings  $\hat{\mu}_i$ , and the shooting parameters  $f_1$  and  $h_1$  in hand, these algebraic conditions recursively fix every remaining coefficient.

In the numerical calculation the integration is not started precisely at the horizon but rather at

$$r = r_0 + \epsilon, \quad 0 < \epsilon \ll 1. \quad (77)$$

The initial data are generated by evaluating the near-horizon series at  $r_0 + \epsilon$ :

$$\begin{aligned} f(r_0 + \epsilon) &= f_1 \left[ \epsilon + f_2 \epsilon^2 + f_3 \epsilon^3 + \dots \right], \\ g(r_0 + \epsilon) &= g_1 \epsilon + g_2 \epsilon^2 + g_3 \epsilon^3 + \dots, \\ h(r_0 + \epsilon) &= f_1^{1/2} \left[ h_1 \epsilon + h_2 \epsilon^2 + h_3 \epsilon^3 + \dots \right], \\ j(r_0 + \epsilon) &= h'(r_0 + \epsilon) = f_1^{1/2} \left[ h_1 + 2h_2 \epsilon + 3h_3 \epsilon^2 + \dots \right]. \end{aligned} \quad (78)$$

These initial values are then fed into the first-order system described in Sec. 5.3. The parameters  $f_1$  and  $h_1$  are adjusted until the solution obeys

$$\lim_{r \rightarrow \infty} f(r) = 1, \quad \lim_{r \rightarrow \infty} g(r) = 1, \quad \lim_{r \rightarrow \infty} h(r) = 1. \quad (79)$$

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